

What is a Network?

A network is a group of computers and other devices that are linked together. There are two kinds of networks that are common. **Local Area Networks (LANs)** are computers that are close together, often in the same room or building. They are connected directly to each other by cables, or **wirelessly** connected through local wi-fi or short-range **wireless** technologies like Bluetooth. Other networks are called **Wide Area Networks (WANs)**. WANs can connect computers in multiple places, even places that are far away from each other. LANs might be connected together through a large WAN. For example, the office of a small company can be connected through a LAN. Each employee's computer could be connected directly through **cables**. Other devices, like printers, large screens, or special equipment, can also be connected to the network. That way, anyone in the network can use those devices.

Many networks are set up as **client-server** systems. A server is a special computer that stores data, or may have other special functions. Client programs ask servers for data, but do not store or manage the data themselves. For example, when you go to a website online, your computer (the client) asks the server for the website data. You can then access the website data on your computer, even though the data is not downloaded onto your computer.

There are different ways to set up a network. A **star network** connects each computer or device to a central server. All connections cross through the central server, making a star shape. **Ring networks** connect one computer to two others, so that all computers are connected in a circle shape. In a **bus network**, each computer is connected to a central cable, called a 'bus'. Lastly, a **mesh network** connects many devices to many others in a complicated system of connections.

Different networks have **advantages** and **disadvantages**. A mesh network is difficult to break since it has so many connections. If one connection has a problem, data can easily find a different route to take. But this is also true for attacks. A virus can infect one computer in a mesh network and easily find a route to other computers. Mesh networks are also complicated and expensive because they involve connecting many cables or high traffic from many wireless devices.

Star networks, on the other hand, are common and easy to set up. Each computer or device connects to a central server. However, if the server has a problem, the entire network stops working. Bus networks have similar advantages and disadvantages. It is easy to connect devices to the central bus, but if the bus has a problem, the entire network stops until it is fixed. In a ring network, data can flow smoothly around the circle of computers, so the network works well under a heavy load. But if one device or cable fails, it can stop the whole network until the problem is fixed, and it can be difficult to add new devices to a ring.

So, how about the internet? What kind of network is the internet? The internet is indeed a network, but it is so big, it is difficult for the human mind to really understand just how the internet is connected. The internet is a massive, global network of interconnected computers and devices, often described as a "network of networks". It links millions of private, public, academic, and government networks together using standardized protocols (TCP/IP) and **physical** infrastructure.

Because the user experience of networks is usually **digital**, through a screen, we sometimes forget that networks are physical. They are built on physical infrastructure, like fiber optic cables, routers, satellites, and data centers. Building and operating a network uses resources, such as electricity, rare

metals, plastics, and water. Because networks are physical things that exist in the world, we need to build them to be efficient, safe, and environmentally friendly.

KEY VOCABULARY

Learn all of these words before you take the quiz for Passage 1.

Local Area Networks (LANs)

Wide Area Networks (WANs)

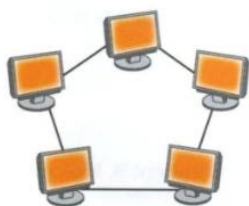
wired / wireless

server / client

physical / digital

advantages / disadvantages

The reading mentioned four different ways to set up a network. Read paragraph 2 again, then match the picture to the set-up.



1 _____



2 _____



3 _____



4 _____